Ashish Shajan

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Profile

Robotics & Automation Engineer with strong hands-on experience in AI-driven systems, robotic manipulation, and autonomous navigation. Specialized in ROS 2, SLAM, PX4, MAVROS, reinforcement learning, and sensor fusion. Experienced in simulation tools such as CARLA, NVIDIA Isaac Sim, Gazebo, and IPG CarMaker. Skilled in delivering autonomous robotics solutions bridging simulation and real-world deployment..

Professional Experience

Chief Technical Officer, Technical Officer — Kerala, India

(AT Bots) *2024 – Present*

- Architected a ROS 2-based autonomous drone stack integrating PX4 and MAVROS for real-time flight control and navigation.
- Designed and implemented multi-sensor fusion (IMU, GNSS, LiDAR) pipelines for accurate localization and obstacle avoidance.
- Directed system-level validation, simulation testing (Gazebo/PyBullet), and hardware deployment for AI-powered structural inspection solutions.

Captain, Autonomous Head

(Team Equinox, SAE ABAJA) 2025 – Present

- Leading the end-to-end development of a Level 2 autonomous BAJA vehicle built from the ground up, driving technical roadmap, milestone planning, and competition strategy.
- Coordinating cross-functional teams (mechanical, electrical, embedded, AI) to ensure seamless hardware–software integration and track readiness.
- Architecting a modular ROS 2-based autonomous stack integrating perception, sensor fusion (Radar, IMU, wheel encoders), localization (EKF), and control modules (ACC, AEB, LKA).
- Overseeing hardware interface development and system-level validation, including simulation workflows using Gazebo, CARLA, and IPG CarMaker prior to real-world deployment.
- Managing real-time integration, testing, and performance optimization to ensure robust autonomous behavior under dynamic track conditions.

Research Intern, — Hyderabad, India

(Robotics Research Centre, IIIT Hyderabad) *2024*

- Developed high-fidelity Gazebo simulation environments to validate autonomous drone control strategies prior to hardware testing.
- Engineered a custom autonomous mobile robot from scratch, implementing 2D LiDAR-based SLAM algorithms to generate high-resolution occupancy grid maps.
- Developed real-time differential drive kinematics and wheel odometry estimation pipelines for precise localization.
- Performed Hardware-in-the-Loop (HITL) testing and PID tuning to improve stability and tracking performance.

Intern - INTEL Unnati Industrial Training Program, — 2024

(INTEL Corporation Bangalore)

- Selected for Intel's national-level industrial training program focused on advanced AI and data-driven system development.
- Worked on knowledge representation and structured dataset analysis to extract actionable insights using machine learning techniques.

• Designed and implemented data preprocessing pipelines, feature engineering workflows, and model validation strategies to improve predictive performance.
• Applied AI concepts to real-world problem statements, emphasizing optimization, accuracy improvement, and scalable deployment.
• Collaborated in a team-based environment, presenting technical findings and solution strategies through structured reports and demonstrations.

Projects

Crack Detection with drones using AI, Academic Project
(PyTorch, ROS2, PX4) *2024*

• Spearheaded and architected a deep learning–based crack detection model optimized for deployment on resource-constrained edge devices, balancing inference latency and detection accuracy for real-world inspections.
• Led end-to-end system design and integration, combining autonomous drone navigation (PX4 + MAVROS + ROS 2) with real-time computer vision pipelines for automated structural inspection.
• Directed cross-functional teams across AI, embedded systems, and flight-control domains, ensuring stable hardware–software integration and successful real-world deployment.

BAJA SAE India 2025: Autonomous Software Stack, Competition Project
(Ros2, Control Systems, Computer Vision) *2025*

• Led the complete autonomous system development for a Level 2 BAJA vehicle, defining architecture for perception, localization, planning, and control modules.
• Designed and implemented radar-based perception and multi-sensor fusion (IMU, wheel encoders) using Extended Kalman Filter for robust real-time localization.
• Developed and integrated ADAS functionalities including Adaptive Cruise Control (ACC), Autonomous Emergency Braking (AEB), and Lane Keep Assist (LKA) within a ROS 2 framework.
• Oversaw simulation, validation, and performance optimization using Gazebo, CARLA, and IPG CarMaker prior to real-world track deployment.
• Coordinated hardware–software integration through CAN-based vehicle interface nodes to ensure stable low-level control execution.

Quadruped Robot (ROS 2, Isaac Sim) Academic Project
(ROS 2, Kinematics, Gazebo, Dynamics) *2024*

- Designed and developed a ROS 2-based locomotion framework for a 12-DOF quadruped robot, implementing modular node architecture for gait control and joint coordination.
- Implemented forward and inverse kinematics algorithms to achieve precise foot trajectory planning and stable stance control.
- Developed and optimized gait patterns (crawl/trot) with dynamic balance control and real-time joint synchronization using ROS 2 Control.
- Integrated IMU-based feedback for orientation stabilization and posture correction during uneven terrain traversal.
- Conducted high-fidelity simulation and physics validation in NVIDIA Isaac Sim and Gazebo to analyze joint torques, stability margins, and dynamic performance before hardware deployment.
- Tuned PID-based joint controllers for smooth motion execution, minimizing oscillations and improving energy efficiency.

Differential Drive Mobile Robot, Research Internship, IIIT Hyderabad
(ROS 2, Raspberry Pi, RPLiDAR) *2025*

- Implemented 2D LiDAR-based SLAM using Cartographer to generate high-resolution occupancy grid maps in real-world environments.
- Developed real-time differential drive kinematics and wheel odometry estimation pipelines.
- Validated SLAM mapping performance in structured indoor environments.

MINI PROJECTS: - Waste segregation with ML, Manipulator and self balancing robot with Roborio, Underwater surveillance system

Skills

- Robotics & Navigation: Autonomous Vehicles, Control Systems, Inverse Kinematics, Autonomous Mobile Robots, Differential Drive Kinematics, Nav2, SLAM (LiDAR), Localization, Odometry, Sensor Fusion (IMU+Encoders)
- AI & Machine Learning: Deep Learning, Computer Vision, PyTorch, YOLOv8, U-Net++, EfficientNet, Model Optimization
- Frameworks & Middleware: ROS 2, PX4 Autopilot, Gazebo, RViz, micro-ROS
- Softwares: Gazebo, Isaac Sim, IPG Carmaker, Rviz, Carla, Pybullet, Matlab, QGroundControl
- Embedded Systems: ESP32, Raspberry Pi, Teensy, PID Control, Trajectory Generation
- Programming & Tools: Python, C++, Embedded C, Linux, Git, Docker, PlatformIO
- Soft Skills - Technical Leadership, Team Collaboration, System-Level Thinking, Strategic Problem-Solving, Decision-Making, Clear Communication, Adaptability, and Rapid Execution.

Education

B.Tech in Robotics and Automation, Kottayam, Kerala *(Saintgits College of Engineering)* *Expected 2026*
Relevant Courses: Control Systems, Robot Kinematics & Dynamics, AI, Machine Learning, Embedded Systems, Computer Vision, Mobile Robots, Robot Operating System.

Higher Secondary in Jawahar Navodaya Vidyalaya Wayanad

Certifications & Achievements

- Intel Unnati Industrial Training – 2 month Internship program
- 3rd Place, Vruthi 2025: Clean Kerala Conclave Wasteathon (Awarded Rs. 25,000 for AI-powered waste segregation).
- 3rd Place, National Hackathon: Recognized for innovative robotics solutions at Rajagiri School of Engineering.
- NPTEL Certification: Soft Skill Development

Languages

- English [Professional Proficiency]
- Malayalam [Native]
- Hindi [Intermediate]